# Middleware Architectures 2

## Lecture 8: Scheduling, Scalability and HPA

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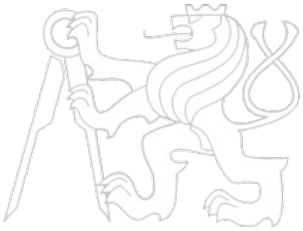

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## Overview

- Kubernetes Scheduling
- Scalability and HPA

# The Kubernetes Scheduler

- The **kube-scheduler** is a control-plane component responsible for Pod placement
  - *Watches for newly created Pods with no assigned node*
  - *Selects the best node for each unscheduled Pod and writes the binding*
  - *Does not start or run the Pod – done by the kubelet on the node*
- Scheduling pipeline runs in two phases
  - ***Filtering*** *– eliminates nodes that cannot run the Pod (hard constraints)*
  - ***Scoring*** *– ranks remaining nodes and picks the highest-score one*
- Scheduler is pluggable via the **Scheduling Framework**
  - *Extension points:* `PreFilter`, `Filter`, `PostFilter`, `PreScore`, `Score`, `Reserve`, `Bind`
  - *Custom plugins can be compiled in or loaded as out-of-tree schedulers*
- Multiple scheduler profiles can coexist in a single cluster



# Filtering – Hard Constraints

- A node passes filtering only when *all* predicates are satisfied
  - ***Resource fit*** *– requested CPU/memory must fit into allocatable capacity*
  - ***Node selector*** *– Pod's* `nodeSelector` *labels must match node labels*
  - ***Node affinity*** *–* `requiredDuringSchedulingIgnoredDuringExecution` *rules*
  - ***Taints and tolerations*** *– Pod must tolerate all taints on the node*
  - ***Volume topology*** *– PVCs must be satisfiable on the candidate node*
  - ***Pod anti-affinity*** *– required rules must not be violated*
- Example: scheduling a GPU workload

```yaml
spec:
  nodeSelector:
    accelerator: nvidia-tesla-a100
  tolerations:
  - key: "gpu"
    operator: "Exists"
    effect: "NoSchedule"
  containers:
  - name: trainer
    image: pytorch/pytorch:2.3
    resources:
      limits:
        nvidia.com/gpu: "1"
```

- Pods that cannot be scheduled remain in `Pending` state

# Scoring – Soft Preferences

- After filtering, each remaining node receives a score 0–100
  - *Final node score is a weighted sum across all enabled scoring plugins*
- Final score formula
  - `finalScore(node) = ∑ pluginScore(node, p) × pluginWeight(p)`
  - *The node with the highest* `finalScore` *is selected; ties are broken randomly*
- Key scoring plugins
  - ***LeastAllocated*** *– prefers nodes with most free CPU and memory*
  - ***MostAllocated*** *– bin-packing; fills nodes before using new ones*
  - ***NodeAffinity*** *– rewards nodes matching preferred affinity rules*
  - ***InterPodAffinity*** *– rewards co-location or spread based on soft rules*
  - ***ImageLocality*** *– prefers nodes that already have the container image*
  - ***TaintToleration*** *– penalises nodes with tolerated-but-present taints*



# Kubernetes Lease-Based Leader Election

- Each controller that needs a leader creates a `coordination.k8s.io/Lease` object
  - *The holder writes its identity and a* ***renewTime*** *timestamp periodically*
  - *If renewTime is stale by more than* `leaseDuration`*, a candidate can acquire the lease*
  - *Optimistic concurrency (resource version) prevents two candidates from acquiring simultaneously*
- Parameters
  - `leaseDuration` *– how long a lease is valid without renewal (default 15 s)*
  - `renewDeadline` *– leader must renew within this window (default 10 s)*
  - `retryPeriod` *– how often candidates retry acquisition (default 2 s)*
- Illustrated state machine

```
Candidate A
  GET Lease → leaseDuration expired?
    YES → PUT Lease(holder=A, renewTime=now)
          → success (no conflict) → LEADER
          → 409 conflict (B won) → wait+retry
  As LEADER: every renewDeadline/2 renew lease
  On crash: lease expires → B or C takes over
```

## Overview

- Kubernetes Scheduling
- Scalability and HPA



## Horizontal vs. Vertical Scaling

- **Vertical scaling (scale-up)** – give the same instance more CPU/memory
  - *Simple to implement; no application changes required*
  - *Pod needs to be restarted (in-place resize is possible)*
- **Horizontal scaling (scale-out)** – add more replicas of the same component
  - *Near-linear throughput increase for stateless services*
  - *Fault tolerance improves with replica count*
  - *Requires applications to be **stateless** or use shared state (DB, cache)*
- Comparison at a glance
  - *Vertical: fast to apply, limited headroom, downtime risk, good for stateful databases*
  - *Horizontal: unlimited (in theory), zero-downtime, preferred for web/API tiers*
- Kubernetes supports both: **HPA** (horizontal) and **VPA** (vertical)

# HPA Overview

- The **HorizontalPodAutoscaler** automatically adjusts the replica count of a workload
  - *Targets:* `Deployment`, `ReplicaSet`, `StatefulSet`, *or any resource implementing the scale subresource*
  - *Operates as a control loop running every 15 s (configurable)*
- Supported metric types
  - ***Resource metrics*** *– CPU/memory utilisation from* `metrics-server`
  - ***Custom metrics*** *– application-specific metrics via* `custom.metrics.k8s.io` *API*
  - ***Object metrics*** *– a single metric on a specific Kubernetes object (e.g. Ingress RPS)*
- HPA does *not* start new nodes
  - *Cluster Autoscaler is needed when no node has capacity*



# HPA Control Loop

- The HPA controller computes the desired replica count using the formula
  - `desiredReplicas = ceil(currentReplicas × (currentMetric / desiredMetric))`
  - *Example: 3 replicas, CPU at 80%, target 50% → ceil(3 × 80/50) = ceil(4.8) =* ***5***
- Scale-down is stabilised to prevent flapping
  - *Default stabilisation window:* ***5 minutes*** *before scaling down*
  - *Scale-up is immediate (no default stabilisation)*
- Control flow

```
Every 15 s:
1. Fetch metric values for all Pods in the target
2. Discard Pods in terminating or not-yet-ready state
3. Compute desiredReplicas per metric; take the maximum
4. if desiredReplicas < minReplicas → use minReplicas
   if desiredReplicas > maxReplicas → use maxReplicas
5. If stabilisation window allows → apply scale
6. Update HPA status (currentReplicas, desiredReplicas, conditions)
```

- Observe current state: `kubectl get hpa -w`
- Detailed events: `kubectl describe hpa <name>`

# HPA Configuration Example

- HPA v2 API allows multiple metrics and fine-grained scaling behaviour

```yaml
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: api-server-hpa
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: api-server
  minReplicas: 2
  maxReplicas: 20
  metrics:
  - type: Resource
    resource:
      name: cpu
      target:
        type: Utilization
        averageUtilization: 60
  - type: Resource
    resource:
      name: memory
      target:
        type: AverageValue
        averageValue: 512Mi
  behavior:
    scaleDown:
      stabilizationWindowSeconds: 300
      policies:
      - type: Percent
        value: 25
        periodSeconds: 60
    scaleUp:
      policies:
      - type: Pods
        value: 4
        periodSeconds: 30
```



# Custom Metrics and KEDA

- Built-in CPU/memory metrics are insufficient for event-driven workloads
- **Custom Metrics API** – adapters expose application metrics to HPA
  - *Popular adapters: Prometheus Adapter, Datadog Cluster Agent*
  - *Example: scale on HTTP request rate from Prometheus*

```yaml
# HPA using a custom Prometheus metric
metrics:
- type: Pods
pods:
metric:
    name: http_requests_per_second
target:
    type: AverageValue
    averageValue: "500"
```

- **KEDA** (Kubernetes Event-Driven Autoscaler)
  - *Extends HPA with 60+ built-in scalers*
  - *Scales on RabbitMQ queue length, Kafka consumer lag, ...*